TX-OMICS REVISION

Seven gene sets.
One auditable curation.

A concise record of the publication rules, pathway selection, identity checks, and overlap that produced the stocker-ready inputs.

7 stocker CSVs

2 analysis families

4 primary sets

3 GO term sets

Two curation families—analyzed separately

Primary evidence sets and CSW-derived GO term-grouped sets use separate overlap analyses.

FAMILY 1 · PRIMARY CURATION SETS

01
6 genes
Mechanistic
Consistent/correlated + published sleep/wake effect

02
20 genes
Homeostatic genes
Opposing history ↔ rebound correlations: 15 + 5

03
97 genes
CSW 4+ genes
Consistent in ≥4 of 7 datasets

04
4 genes
HLH genes
Potential upstream regulators

FAMILY 2 · CSW-DERIVED GO TERM-GROUPED SETS (FC0.5 / FC1)

05
99 genes
Ribosomal
DAVID term hits; approved keywords

06
167 genes
Mitochondrial
DAVID mito/metabolism term hits

07
5 genes
Immune
DAVID immune / Toll / Imd term hits

RULE
Keep them separate
Report overlap within each family—never as one seven-set comparison.

Two curation tracks converge on one controlled folder

They share identity gates and file standards—but their overlap analyses remain separate.

FAMILY 1

Primary curation sets

1. Paper-defined evidence
2. Apply four distinct rules
3. Resolve FlyBase identity
4. **Publish sets 01-04**

FAMILY 2

GO term-grouped sets

1. CSW FC0.5 + FC1 tables
2. DAVID · FDR ≤10%
3. Review term keywords
4. **Publish sets 05-07**

902 source observations

791 unique source FBgns

≥2 minimum pathway frequency

7 experiments

Thresholds remain separate: FC0.5 and FC1 are never collapsed into a max-frequency field.

Mechanistic and Homeostatic are different evidence categories

The six-gene functional set is distinct from the 20-gene opposing history–rebound correlation set.

SET 01 · 6 GENES

Mechanistic

Consistent or correlated sleep/wake genes with published effects in a direction consistent with potential homeostatic function.

- Expression is consistent across conditions or correlated with sleep/wake history.
- Published perturbation affects sleep/wake in a direction compatible with negative feedback.
- The paper highlights six such genes.

unc79 · SIFa · rumpel · AstA-R2 · Trhn · RpL23

SET 02 · 20 GENES

Homeostatic genes

15

History+ / Rebound−

5

History− / Rebound+

Genes correlate with sleep/wake history before sampling and with rebound in the opposite direction afterward.

Identity correction

Source row trh / FBgn0262139 is resolved from publication context to Trhn / FBgn0035187 (tryptophan hydroxylase). The wrong FBgn is never retained.

CSW consistency and HLH regulation are separate categories

Set 03 measures recurrence across seven datasets; Set 04 identifies potential upstream regulators.

SET 03 · 97 GENES

CSW 4+ genes

97
FC0.5 wake frequency 4–6

7
also qualify at FC1

0
sleep genes at frequency 4

Category rule

- Consistent sleep/wake genes evident across four or more of seven datasets.
- The current output contains 97 FC0.5 wake genes; seven also qualify at FC1.
- No sleep-direction gene reaches frequency 4 in this run.

SET 04 · 4 GENES

HLH genes

Named in Results / Table S1 after HOMER identified a CAGCTG E-box upstream of wake-induced genes.

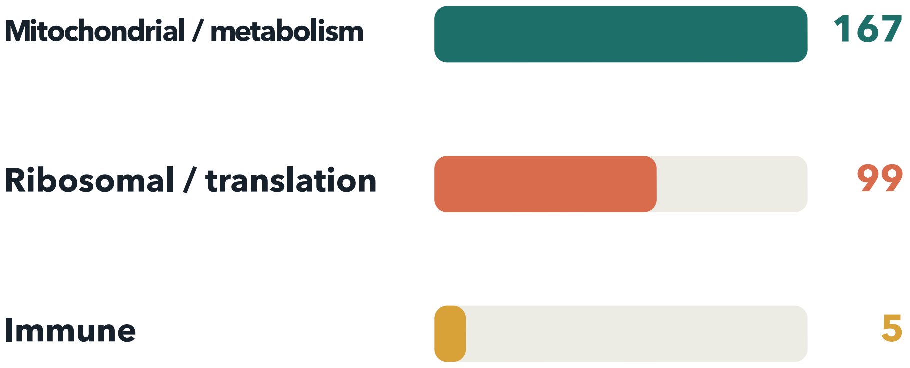
bigmax	FBgn0039509
HLH3B	FBgn0011276
E(spl)m3-HLH	FBgn0002609
E(spl)mbeta-HLH	FBgn0002733

CSW-derived GO term-grouped genes retain threshold provenance

Each gene records its source as FC0.5, FC1, or both; only hits from approved FDR-passing terms enter sets 05–07.

1. Submit four CSW masters
2. Keep FDR ≤10% terms
3. Review term keywords
4. Take hit FBgns

GO term-grouped outputs



All three outputs are wake-direction genes.

Threshold provenance · 240 unique FBgns

203	**FC0.5 only**	Ribosomal and mitochondrial dominate
5	**FC1 only**	BomBc2 · BomS1 · BomS2 · IM14 · Thor
32	**both thresholds**	FC1 ribosomal/mitochondrial genes are also at FC0.5

Key takeaway **Only the five immune genes are FC1-exclusive.**

Identity resolution is conservative by design

Current symbols can pass directly; ambiguous synonyms require biological context or a case-supported source match.

PASS	**Trhn**	FBgn0035187	Current symbol · tryptophan hydroxylase neuronal · publication-resolved in Homeostatic.
REJECT	**trh**	FBgn0262139	Current symbol · trachealess · must not enter any generated identity field for this publication row.
REVIEW	**Trh**	ambiguous	Synonym of Hn, Trhn, and trh · never auto-approved without biological context.

Approval gates

- Exact current symbol
- Publication-supported correction
- Unique case-insensitive current-symbol match
- Otherwise: stop for review

Membership key

flybase_gene_id

Symbols are display labels. All overlap, deduplication, and set membership use FBgn only.

Primary-set overlap is limited to three genes

Mechanistic, Homeostatic genes, CSW 4+ genes, and HLH genes are compared only with one another.

6	20	97	4
Mechanistic	Homeostatic genes	CSW 4+ genes	HLH genes

Observed shared genes

Mechanistic × Homeostatic genes	2	Trhn · AstA-R2
Mechanistic × CSW 4+ genes	1	RpL23
All other pairings	0	No shared FBgns

Within Family 1

124	unique FBgns
121	in exactly one set
3	in exactly two sets
0	in three or four sets

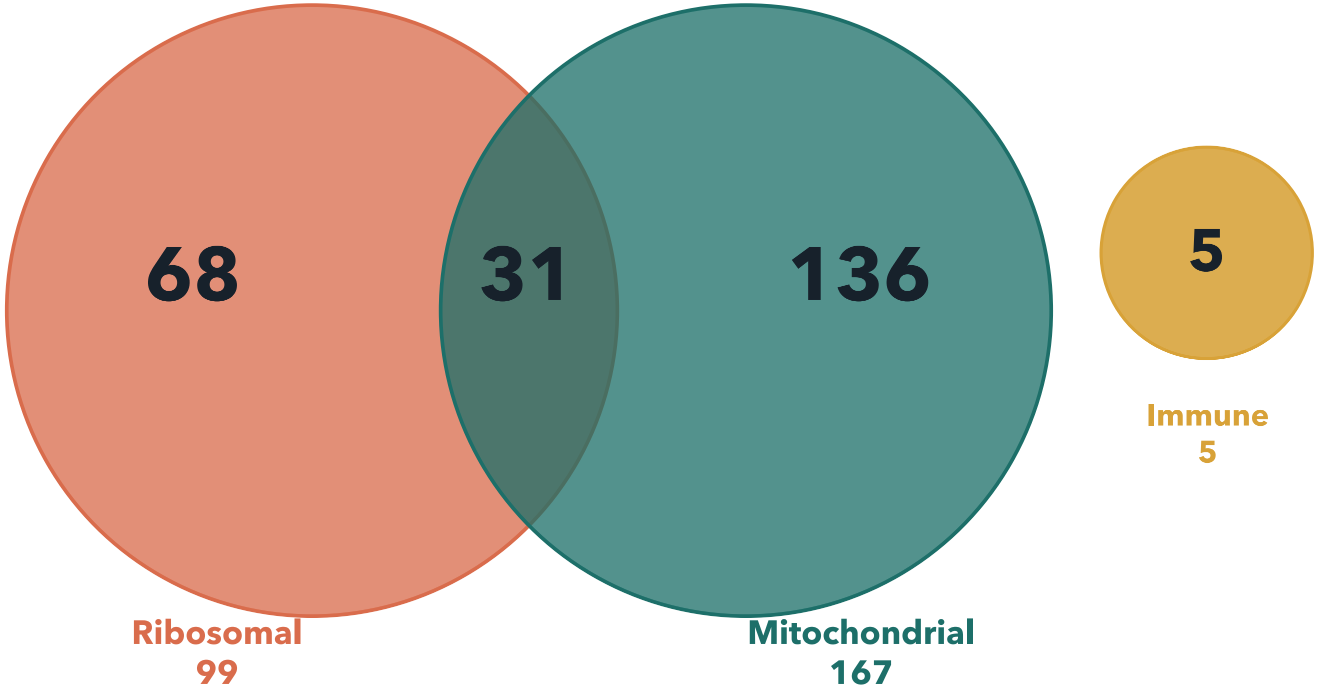
HLH is fully disjoint from the other primary sets.

GO term-grouped overlap is confined to ribosomal × mitochondrial

The three pathway buckets are compared only within Family 2; threshold provenance remains a separate attribute.

GO term-group membership

Schematic · counts shown



240 unique FBgns in Family 2

What the overlap means

31 genes occur in both ribosomal and mitochondrial buckets

5 immune genes are fully disjoint

0 immune overlaps with either other GO bucket

The 31-gene overlap is intentional: mitochondrial translation terms match both keyword groups.

No cross-family overlap is shown: Family 1 and Family 2 answer different curation questions.

The contract is simple—and strict.

7 Exactly seven CSVs in the discovery folder

2 Required keys: ext_gene and flybase_gene_id

0 No overlap, GO, or audit CSVs beside the inputs

Seven inputs

- 01 Mechanistic
- 02 Homeostatic genes
- 03 CSW 4+ genes
- 04 HLH genes
- 05 Ribosomal
- 06 Mitochondrial
- 07 Immune

RUN

```
python -m fl_ai_reagent_stocker run ./data/gene_sets/Tx-Omics_Revision/ \
  --config ./data/config/stock_split_config_priority_example.json
```

Stability rule Discovery is recursive: every extra *.csv changes the run. Keep analysis outputs elsewhere.